

ML LAB EXPT – 1

Aim of the Experiment

Develop a program to create histograms for all numerical features and analyze the distribution of each feature. Generate box plots for all numerical features and identify any outliers. Use California Housing dataset.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the California Housing dataset
try:
    df = pd.read_csv('california_housing_train.csv')
except FileNotFoundError:
    print("Error: california_housing_train.csv not found.")
    exit()

# Function to create histograms
def create_histograms(dataframe, features):
    for feature in features:
        plt.figure(figsize=(8, 6))
        sns.histplot(dataframe[feature], kde=True) #The Kernel Density
Estimate (KDE) is added to visualize the distribution curve.
        plt.title(f'Histogram of {feature}')
        plt.xlabel(feature)
        plt.ylabel('Frequency')
        plt.show()

# Function to create box plots
def create_boxplots(dataframe, features):
    for feature in features:
        plt.figure(figsize=(8, 6))
        sns.boxplot(x=dataframe[feature])
        plt.title(f'Boxplot of {feature}')
        plt.show()

# Select numerical features (you might need to adjust based on your
dataset)
numerical_features = ['longitude', 'latitude', 'housing_median_age',
'total_rooms', 'total_bedrooms', 'population', 'households',
'median_income', 'median_house_value']
```

```

# Create histograms
create_histograms(df, numerical_features)

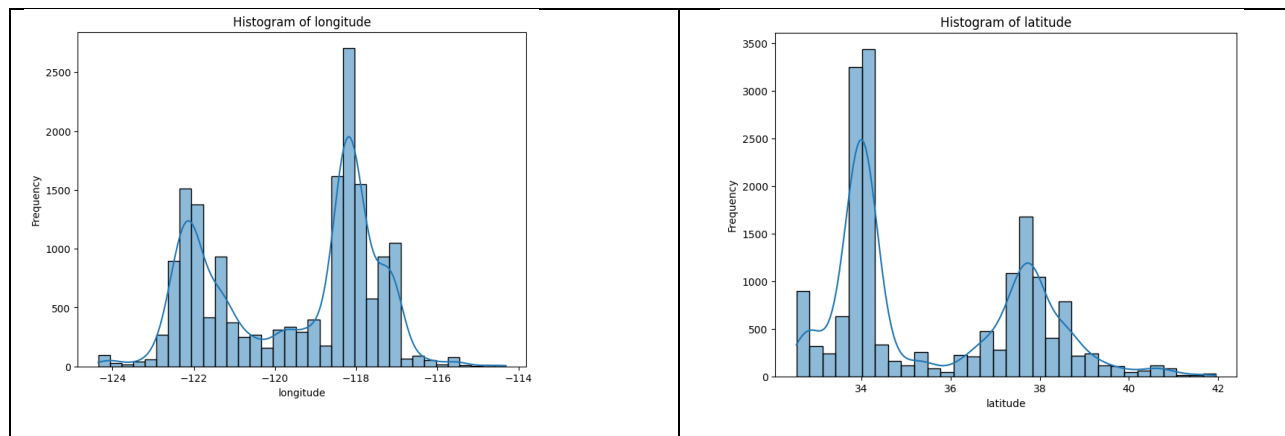
# Create box plots
create_boxplots(df, numerical_features)

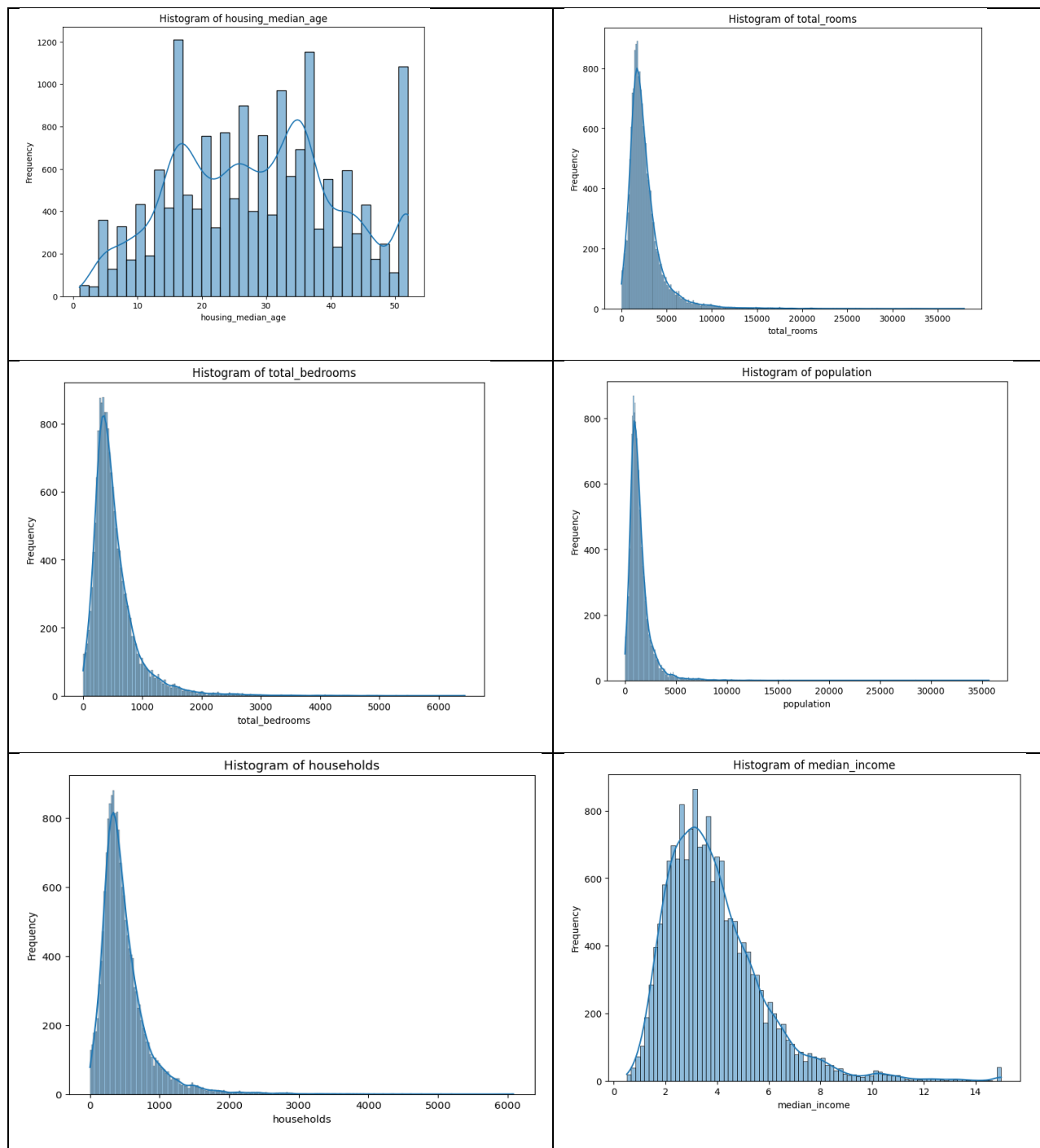
# Analyze distributions and identify outliers using the Interquartile
Range (IQR) method.
print("Analysis:")
for feature in numerical_features:
    q1 = df[feature].quantile(0.25)
    q3 = df[feature].quantile(0.75)
    iqr = q3 - q1
    lower_bound = q1 - 1.5 * iqr
    upper_bound = q3 + 1.5 * iqr
    outliers = df[(df[feature] < lower_bound) | (df[feature] > upper_bound)]

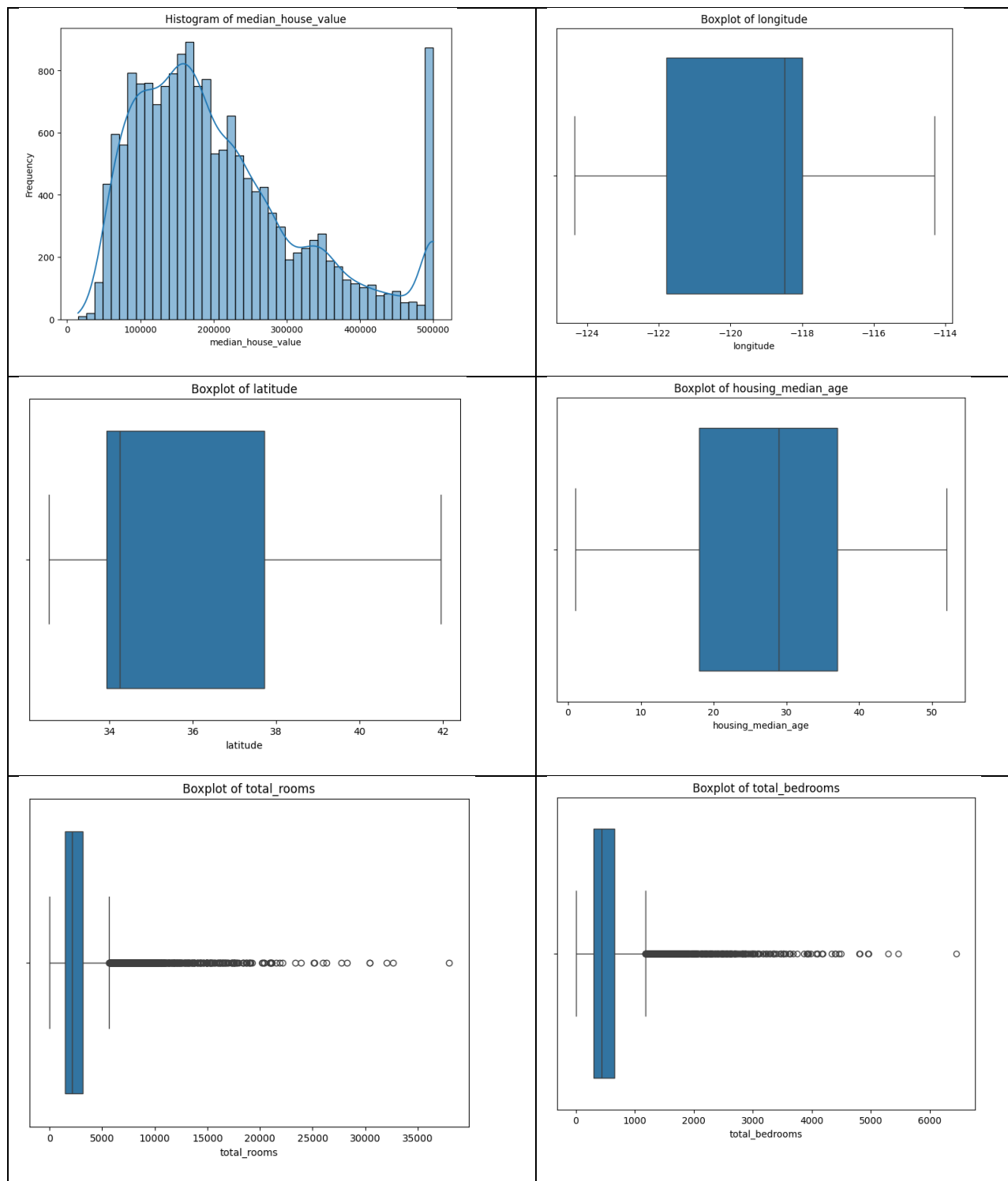
    print(f"\nFeature: {feature}")
    if not outliers.empty:
        print(f"Number of outliers: {len(outliers)}")
        print("Outlier values:")
        print(outliers[feature])
    else:
        print("No outliers detected.")

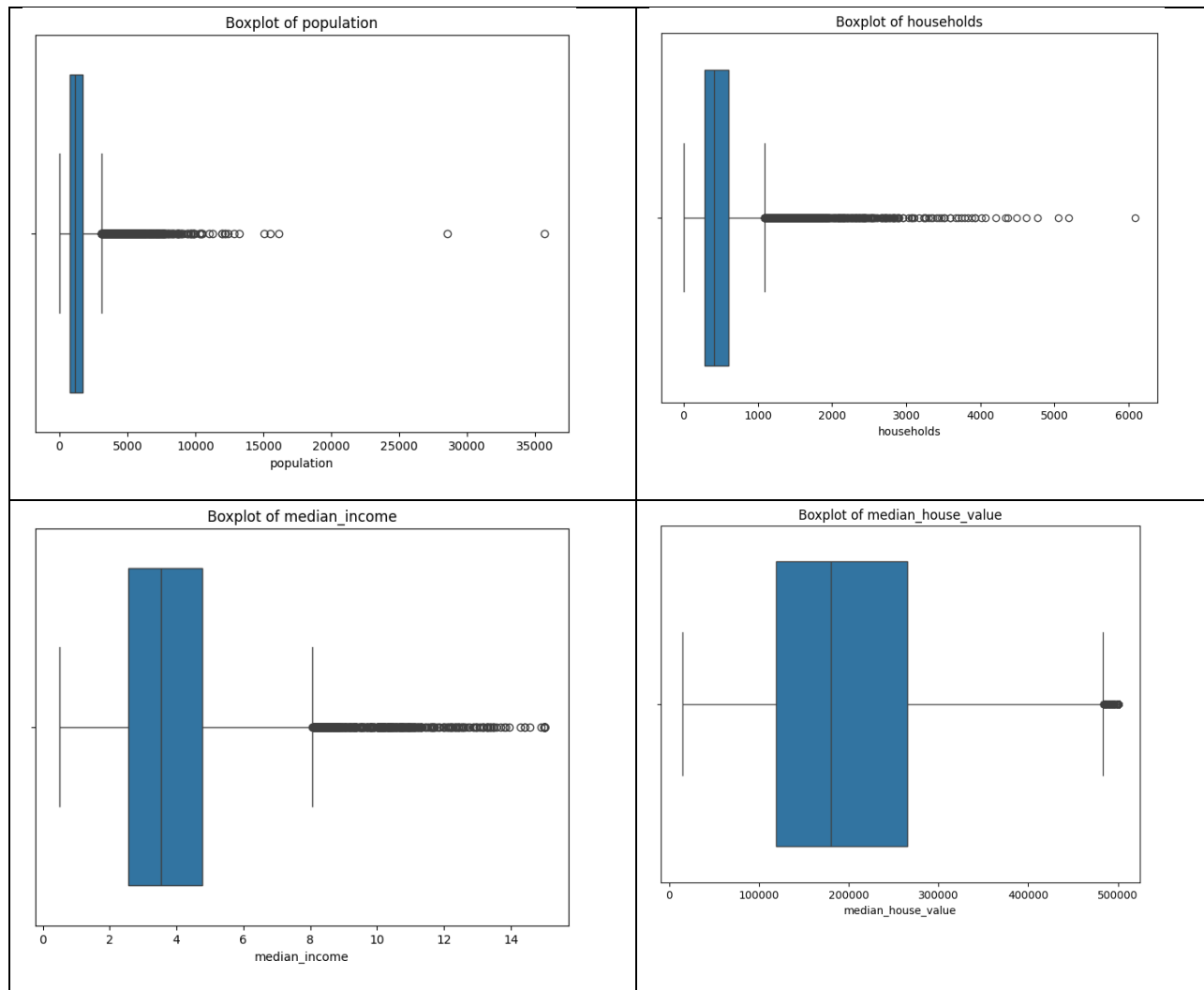
```

Output:









Analysis:

Feature: longitude
No outliers detected.

Feature: latitude
No outliers detected.

Feature: housing_median_age
No outliers detected.

Feature: total_rooms
Number of outliers: 1076

Outlier values:

1	7650.0
133	10377.0
135	9444.0
168	9033.0
172	7437.0

```
...
16652    6719.0
16661    5823.0
16679    6837.0
16692    6190.0
16986    5694.0
Name: total_rooms, Length: 1076, dtype: float64
```

```
Feature: total_bedrooms
Number of outliers: 1074
Outlier values:
```

```
0        1283.0
1        1901.0
115      1257.0
133      2331.0
135      1997.0
```

```
...
16490    1390.0
16546    1606.0
16569    1374.0
16587    1211.0
16679    1417.0
Name: total_bedrooms, Length: 1074, dtype: float64
```

```
Feature: population
Number of outliers: 1015
Outlier values:
```

```
8         3134.0
24        3424.0
44        3565.0
127       4042.0
133       4507.0
```

```
...
16490    3551.0
16546    4323.0
16679    3468.0
16824    3148.0
16949    3546.0
Name: population, Length: 1015, dtype: float64
```

```
Feature: households
Number of outliers: 1032
Outlier values:
```

```
133      1807.0
135      1482.0
164      1186.0
165      1243.0
168      1845.0
```

```
...
16474    1242.0
16490    1392.0
16546    1475.0
16569    1189.0
16679    1405.0
```

Name: households, Length: 1032, dtype: float64

Feature: median_income

Number of outliers: 563

Outlier values:

172	10.2948
480	8.0930
521	8.2531
541	8.4284
646	8.4721

...

16419	10.6272
16420	8.9248
16422	8.6340
16434	9.1834
16512	9.5551

Name: median_income, Length: 563, dtype: float64

Feature: median_house_value

Number of outliers: 895

Outlier values:

215	500001.0
260	500001.0
862	500001.0
1388	500001.0
1404	500001.0

...

16420	500001.0
16421	500001.0
16434	500001.0
16482	493800.0
16512	500001.0

Name: median_house_value, Length: 895, dtype: float64